

AI-INTEGRATED AIR POLLUTION FORECASTING AND POLICY DASHBOARD FOR DELHI-NCR

ABSTRACT

Air pollution in Delhi-NCR has become a persistent public health and environmental challenge, driven by vehicular emissions, industrial discharge, and seasonal activities like crop residue burning. Despite the availability of extensive air quality datasets, existing forecasting systems mainly focus on short-term pollutant prediction and fail to provide actionable insights regarding specific pollution sources. Consequently, policymakers and citizens are unable to take preventive actions. The affected community includes millions of residents across Delhi, Gurugram, Noida, and Ghaziabad, where poor air quality regularly exceeds safe limits, leading to severe respiratory and cardiovascular issues. Previous studies using machine learning algorithms have shown success in pollutant forecasting but suffer from overfitting, missing data and limited source attribution. To address these issues, the proposed system integrates an AI-driven framework combining hybrid deep learning and data fusion from ground sensors, meteorological datasets, and satellite imagery. The model performs both pollution level forecasting and probabilistic source identification through clustering techniques. A real-time web-based dashboard visualizes pollutant trends and probable sources. The system includes automated data cleaning, adaptive retraining and gives results that are human understandable that builds people trust and for improved reliability, transparency. The project thus aims to help decision-makers with an easy-to-understand tool that provides useful insights and suggestions to lower pollution in the Delhi-NCR region.

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